

Anomaly Detection for Fraud and Sensor Data

Summer Internship Project Report 2026

Student Name: Shreetama Dey

Course: B.Sc. Statistics (Hons.)

Batch: 2024 – 2028

Institute: St. Xavier's College (Autonomous), Kolkata

Period of Internship: 18th May, 2026 – 14th June, 2026

Report submitted to: IDEAS – Institute of Data Engineering,
Analytics and Science Foundation, ISI Kolkata

Abstract

This project investigates anomaly detection in credit card transactions using statistical and machine learning techniques. A synthetic dataset of 1,000 credit card transactions was generated that included 950 normal transactions and 50 fraudulent ones. Four anomaly detection techniques were implemented and compared: Z-Score method using distance from home feature, Interquartile Range (IQR) method using transaction amount, Isolation Forest algorithm and Local Outlier Factor (LOF) algorithm. After preprocessing and scaling the data, model performance was evaluated using precision, recall and confusion matrices. Results demonstrated that Isolation Forest achieved perfect classification with 100% precision and recall, while LOF showed moderate performance with 36% precision and recall. The project illustrates the effectiveness of multivariate machine learning techniques over univariate statistical approaches for fraud detection problems.

Contents

1	Introduction	3
2	Project Objectives	4
3	Methodology	5
3.1	Data Generation	5
3.2	Data Preprocessing	6
3.3	Exploratory Data Analysis	6
3.4	Anomaly Detection Methods	6
3.4.1	Z-Score Method	6
3.4.2	Interquartile Range (IQR) Method	6
3.4.3	Isolation Forest	7
3.4.4	Local Outlier Factor (LOF)	7
3.5	Model Performance Evaluation	8
3.6	Workflow Diagram	9
4	Data Analysis and Results	10
4.1	Dataset Overview	10
4.2	Descriptive Analysis	10
4.3	Statistical Anomaly Detection Results	10
4.3.1	Z-Score Method	10
4.3.2	Interquartile Range (IQR) Method	11
4.4	Machine Learning Anomaly Detection Results	11
4.4.1	Isolation Forest	11
4.4.2	Local Outlier Factor (LOF)	11
4.5	Comparative Analysis of Detection Methods	12
5	Results	13
6	Conclusion	14
7	Appendices	15
7.1	References	15
7.2	GitHub Repository Link for Source Codes	15

1 Introduction

With the growing popularity of online payment systems, the process of financial transactions has changed significantly. Credit cards are currently considered one of the easiest and most efficient ways to make payments. Nonetheless, along with the growth in electronic transactions, there has been a significant rise in frauds. Credit card fraud poses significant financial risks to both consumers and financial institutions and is thus of great importance for research in data analytics and finance security. Nevertheless, detecting such anomalies accurately is a challenging task because fraudulent transactions often constitute only a small fraction of the total transaction volume.

Several ways for anomaly detection have been proposed through statistical and machine learning techniques. Traditional statistical approaches such as Z-Score Analysis and the Interquartile Range (IQR) method identify observations that lie far from the central tendency of the data. Although these techniques are easy to understand and apply, they often consider only a single feature at a time and may fail to capture complex relationships among variables. Machine learning-based anomaly detection techniques offer a more flexible approach by analyzing multiple features simultaneously. Isolation Forest identifies anomalies by recursively isolating observations using random partitioning and Local Outlier Factor (LOF) detects anomalies based on local density differences between observations and their neighbors.

In this project, a synthetic credit card transaction dataset was created to simulate realistic fraud scenarios. The objective was to compare the effectiveness of statistical and machine learning approaches in detecting fraudulent transactions and to evaluate their strengths and limitations.

Topics Covered During the First Two Weeks of Internship

- Python Programming Basics
- Machine Learning Fundamentals
- Large Language Model (LLM) Foundations

2 Project Objectives

1. **To generate a Synthetic Credit Card Transaction Dataset:** To create a realistic dataset with 1,000 transactions containing both normal and fraudulent transactions, ensuring class imbalance (95% normal, 5% fraud).
2. **To implement Univariate Statistical Anomaly Detection Methods:** To apply Z-Score method on the distance from home feature and IQR method on the transaction amount feature to identify statistical outliers.
3. **To implement Multivariate Machine Learning Models:** To train Isolation Forest and Local Outlier Factor (LOF) algorithms using multiple features simultaneously to detect anomalous transactions.
4. **To evaluate and compare Model Performance:** To assess each method using precision, recall and confusion matrix metrics to determine the most effective approach for fraud detection.

3 Methodology

3.1 Data Generation

A synthetic credit card transaction dataset was generated using NumPy’s random number generators with a fixed random seed (42) for reproducibility. The dataset contains 1,000 transactions with four features: amount, hour of the day, number of transactions in last 24 hours and distance from home (in km).

The data was generated in such a manner that normal transactions have lower transaction amounts, mostly occur during daytime hours, involve fewer transactions in the last 24 hours and are generally close to the customer’s home location, and fraudulent transactions have higher transaction amounts, are more likely to occur during unusual late-night hours, involve a larger number of transactions within the last 24 hours and occur farther from the customer’s home location.

Table 3.1: Feature Distributions for Normal and Fraudulent Transactions

Feature	Genuine Transactions	Fraudulent Transactions
amount	Normal (mean=50, sd=15)	Normal (mean=300, sd=100)
hour_of_day	Uniform (7 to 22)	Values from 0 to 5 and 22 to 23
transactions_last_24h	Poisson (lambda=2)	Poisson (lambda=10)
distance_from_home_km	Exponential (mean=3)	Exponential (mean=50)
is_fraud	0	1

Table 3.2: Feature Descriptions

Feature	Description
amount	Transaction value
hour_of_day	Hour when transaction occurred (0 to 23)
transactions_last_24h	Number of transactions in past 24 hours
distance_from_home_km	Distance from cardholder’s home in kilometers
is_fraud	Binary indicator variable

3.2 Data Preprocessing

The normal and fraudulent transactions were first combined into a single dataset. The dataset was then randomly shuffled and the indices were reset to ensure that transactions were distributed randomly and that no ordering bias remained. Before applying machine learning models, the feature matrix was standardized using StandardScaler from scikit-learn. Standardization transforms features to have zero mean and unit variance, which is essential for distance-based algorithms like LOF.

3.3 Exploratory Data Analysis

To gain preliminary insights into the characteristics of normal and fraudulent transactions, the dataset was grouped by the target variable (`is_fraud`) and the mean values of key features were computed. This comparison helped identify behavioral differences between the two classes.

3.4 Anomaly Detection Methods

3.4.1 Z-Score Method

The Z-Score method is a statistical technique that measures how far an observation lies from the mean of a distribution in terms of standard deviations. Observations with large absolute Z-scores are considered unusual and may indicate anomalous behavior. In this study, the **distance_from_home_km** variable was used because fraudulent transactions were expected to occur farther from a customer's usual location. Transactions with an absolute Z-score greater than 3 were classified as anomalies.

$$Z = \frac{x - \mu}{\sigma} \quad (3.1)$$

where x is the observation, μ is the mean, and σ is the standard deviation.

3.4.2 Interquartile Range (IQR) Method

The Interquartile Range (IQR) method is a robust statistical approach for detecting outliers. It identifies observations that lie unusually far from the central 50% of the data. In this study, the **transaction amount** variable was analyzed because fraudulent transactions often involve unusually high spending amounts.

$$\text{IQR} = Q_3 - Q_1 \quad (3.2)$$

$$\text{Lower Bound} = Q_1 - 1.5 \times \text{IQR} \quad (3.3)$$

$$\text{Upper Bound} = Q_3 + 1.5 \times \text{IQR} \quad (3.4)$$

Transactions with amounts outside these bounds were classified as anomalies.

3.4.3 Isolation Forest

Isolation Forest is an unsupervised machine learning algorithm specifically designed for anomaly detection. The method isolates observations through repeated random partitioning of the feature space. Since anomalies are rare and different from normal observations, they tend to require fewer partitions to become isolated.

The model was trained using all four transaction features:

- Transaction Amount
- Hour of Day
- Transactions in the Last 24 Hours
- Distance from Home (km)

The Isolation Forest model was implemented with:

- **Contamination = 0.05**: Specifies the expected proportion of anomalies in the dataset. Since the synthetic dataset contained 5% fraudulent transactions (50 out of 1000), the contamination parameter was set to 0.05.
- **Random State = 42**: Fixes the random seed used by the algorithm, ensuring that the results are reproducible across multiple runs.

The algorithm returns a prediction of -1 for anomalous observations and 1 for normal observations. For consistency with the fraud labels in the dataset, these outputs were encoded into binary values, where 1 represents an anomaly and 0 represents a normal transaction.

3.4.4 Local Outlier Factor (LOF)

Local Outlier Factor (LOF) is a density-based anomaly detection algorithm. It evaluates the local density of each observation relative to its nearest neighbors. Observations that have substantially lower density than surrounding points are considered anomalies.

The model was trained on the same four standardized features:

- Transaction Amount
- Hour of Day

- Transactions in the Last 24 Hours
- Distance from Home (km)

The LOF model was implemented with:

- **Number of Neighbors (k) = 20:** Determines the number of nearest neighbors considered when estimating the local density around each observation. A value of 20 allows the model to compare each transaction with a sufficiently large local neighborhood.
- **Contamination = 0.05:** Specifies that approximately 5% of the observations are expected to be anomalous, corresponding to the proportion of fraudulent transactions in the synthetic dataset.

Similar to Isolation Forest, observations identified as anomalies were assigned a value of 1, while normal transactions were assigned a value of 0.

3.5 Model Performance Evaluation

The anomaly detection methods were evaluated by comparing their predicted labels with the true fraud labels in the dataset. Confusion matrices were generated to summarize classification outcomes. Model performance was assessed using precision and recall, defined as:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3.5)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3.6)$$

where TP denotes true positives, FP denotes false positives, and FN denotes false negatives. These metrics were selected because fraud detection involves identifying a small number of anomalous transactions within a largely normal dataset.

3.6 Workflow Diagram

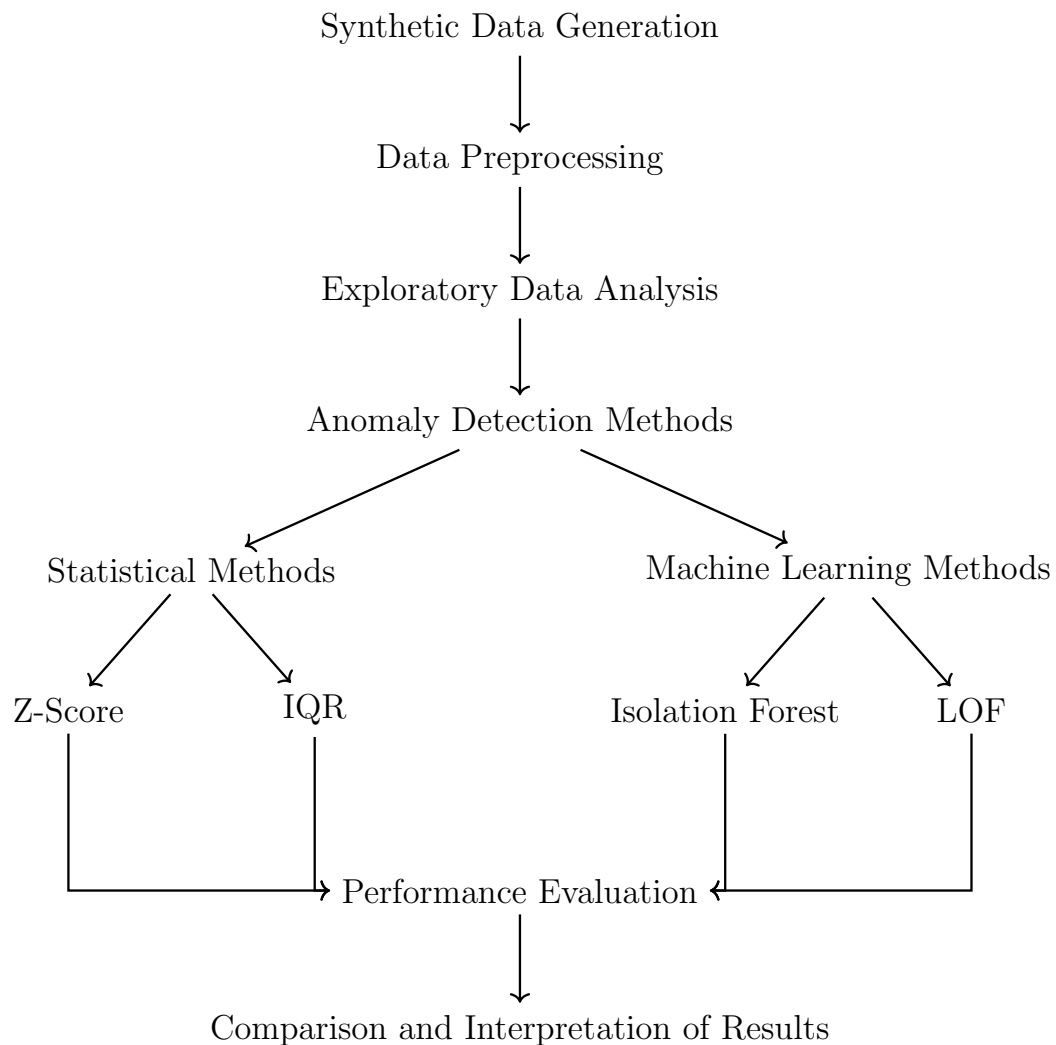


Figure 3.1: Workflow of the anomaly detection study.

GitHub Repository

The complete Python code for this project is available at:

<https://github.com/shreetamadey/Anomaly-Detection-Fraud-Sensor-Data>

4 Data Analysis and Results

4.1 Dataset Overview

The synthetic credit card transaction dataset consists of 1,000 observations and five variables. Among these transactions, 950 (95%) were generated as legitimate transactions and 50 (5%) as fraudulent transactions. This class imbalance closely resembles real-world fraud detection problems, where fraudulent transactions constitute only a small fraction of total transactions.

4.2 Descriptive Analysis

To understand the differences between normal and fraudulent transactions, the mean values of each feature were computed separately for the two classes.

Table 4.1: Mean Feature Values by Transaction Type

Feature	Normal	Fraud
Transaction Amount	50.30	279.37
Hour of Day	13.95	9.72
Transactions in Last 24 Hours	2.02	9.84
Distance from Home (km)	2.96	60.33

The descriptive statistics reveal substantial differences between normal and fraudulent transactions. Fraudulent transactions exhibited substantially higher transaction amounts, greater distances from home and a larger number of transactions within the previous 24 hours compared to normal transactions. Such patterns suggest that these features may be useful indicators for distinguishing fraudulent activity and justify their inclusion in the subsequent anomaly detection analyses.

4.3 Statistical Anomaly Detection Results

4.3.1 Z-Score Method

The Z-Score method was applied to the **distance-from-home** feature. Transactions with an absolute Z-score greater than three were classified as anomalies.

Table 4.2: Z-Score Detection Results

Metric	Value
Anomalies Detected	15

Only 15 transactions were identified as anomalies. Since the method relies solely on a single feature and detects only extreme deviations from the mean, many fraudulent transactions remained undetected. This demonstrates the limitations of univariate anomaly detection techniques in complex fraud detection problems.

4.3.2 Interquartile Range (IQR) Method

The IQR method was applied to **transaction amount**.

Table 4.3: IQR Detection Results

Metric	Value
Anomalies Detected	52

The IQR method identified 52 anomalous transactions. Since fraudulent transactions were generated with substantially larger transaction amounts, the method successfully captured most fraud cases. However, some genuine transactions with unusually large amounts were also flagged as anomalies.

4.4 Machine Learning Anomaly Detection Results

4.4.1 Isolation Forest

The Isolation Forest model was trained using all four standardized features.

Table 4.4: Isolation Forest Confusion Matrix

	Predicted Normal	Predicted Fraud
Actual Normal	950	0
Actual Fraud	0	50

The model achieved perfect classification performance, correctly identifying all fraudulent and legitimate transactions. No false positives or false negatives were observed.

4.4.2 Local Outlier Factor (LOF)

The Local Outlier Factor model was trained using the same feature set.

Table 4.5: LOF Confusion Matrix

	Predicted Normal	Predicted Fraud
Actual Normal	918	32
Actual Fraud	32	18

The LOF model correctly identified 18 fraudulent transactions while failing to detect 32 fraud cases. It also incorrectly classified 32 legitimate transactions as fraudulent. This indicates that LOF struggled to distinguish fraudulent observations from normal observations in the generated dataset.

4.5 Comparative Analysis of Detection Methods

Table 4.6: Comparison of Anomaly Detection Methods

Method	Type	Features Used	Anomalies Detected
Z-Score	Statistical	Distance from Home	15
IQR	Statistical	Transaction Amount	52
Isolation Forest	Machine Learning	All Features	50
LOF	Machine Learning	All Features	50

Table 4.7: Model Performance Comparison for Machine Learning Models

Method	Precision	Recall
Isolation Forest	1.00	1.00
LOF	0.36	0.36

The results demonstrate that multivariate machine learning approaches outperform univariate statistical methods. While the Z-Score and IQR methods rely on a single feature, Isolation Forest and LOF simultaneously consider all transaction characteristics.

Among all methods, Isolation Forest produced the strongest results, achieving perfect fraud detection performance. LOF performed considerably worse despite detecting the same number of anomalies. This is because LOF identifies anomalies based on local density differences, but, the fraudulent transactions in the synthetic dataset exhibited similar characteristics and formed a relatively dense cluster. Consequently, many fraudulent transactions appeared locally normal to the algorithm, leading to lower precision and recall compared to Isolation Forest.

5 Results

The analysis revealed clear differences between genuine and fraudulent transactions. Fraudulent transactions were characterized by significantly higher transaction amounts, greater distances from the cardholder’s home and a larger number of transactions within the previous 24 hours.

Out of all the statistical anomaly detection techniques employed, the IQR method proved to be better than the Z-Score method, indicating that the transaction amount is a more useful predictor for fraud detection as compared to just the distance from home. However, both methods were limited by their reliance on a single variable and were therefore unable to fully capture the multidimensional nature of fraudulent behavior.

The machine learning approaches demonstrated superior performance by simultaneously utilizing all available features. Isolation Forest achieved perfect classification, correctly identifying all fraudulent and genuine transactions without any false positives or false negatives. In contrast, Local Outlier Factor (LOF) achieved lower precision and recall, as the fraudulent transactions formed a relatively dense cluster with similar characteristics. Since LOF detects anomalies based on local density differences, many fraudulent observations appeared locally normal and were therefore not correctly identified.

Overall, the results demonstrate that multivariate machine learning methods are more effective than univariate statistical techniques for detecting fraudulent credit card transactions in the synthetic dataset. Among the methods evaluated, Isolation Forest emerged as the most effective anomaly detection approach.

6 Conclusion

Anomaly detection works as an efficient tool for preventing financial fraud through its ability to detect suspicious transactions before they do any irreversible harm. This project demonstrated the ability to combine statistics and machine learning techniques to help detect abnormalities within transactions. As these models continue to evolve and integrate with real-world banking systems, they pave the way for a safer, more secure and fraud-free financial ecosystem where legitimate transactions flow seamlessly while malicious activities are stopped at the source.

Limitations :

Despite the promising results, the study has certain limitations. The analysis was conducted on a synthetic dataset generated using predefined distributions and assumptions. As a result, the fraudulent transactions may be more distinguishable than those encountered in real-world financial systems. Additionally, only four transaction features were considered, whereas actual fraud detection systems often rely on a much larger set of behavioral and contextual variables. The study also focused on unsupervised anomaly detection methods and did not explore supervised classification techniques that may be available when labeled data exist.

Future Scopes :

Future work can extend this study by applying the proposed methods to real-world credit card transaction datasets containing more complex and evolving fraud patterns. Incorporating richer transaction features, performing optimization and developing real-time fraud detection frameworks could further enhance detection performance. Such extensions would provide a more comprehensive understanding of anomaly detection in practical financial environments.

7 Appendices

7.1 References

1. Liu, Fei Tony Ting, Kai Zhou, Zhi-Hua. (2009). Isolation Forest. 413 - 422. 10.1109/ICDM.2008.17.
2. Breunig, Markus Kröger, Peer Ng, Raymond Sander, Joerg. (2000). LOF: Identifying Density-Based Local Outliers.. ACM Sigmod Record. 29. 93-104. 10.1145/342009.335388.
3. Chandola, Varun Banerjee, Arindam Kumar, Vipin. (2009). Anomaly Detection: A Survey. ACM Comput. Surv.. 41. 10.1145/1541880.1541882.
4. IsolationForest. (n.d.). Scikit-learn. <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.IsolationForest.html>
5. LocalOutlierFactor. (n.d.). Scikit-learn. <https://scikit-learn.org/stable/modules/generated/sklearn.neighbors.LocalOutlierFactor.html>
6. IDEAS TIH Summer Internship Program Materials.

7.2 GitHub Repository Link for Source Codes

The complete Python code for this project is available at:

<https://github.com/shreetamadey/Anomaly-Detection-Fraud-Sensor-Data>